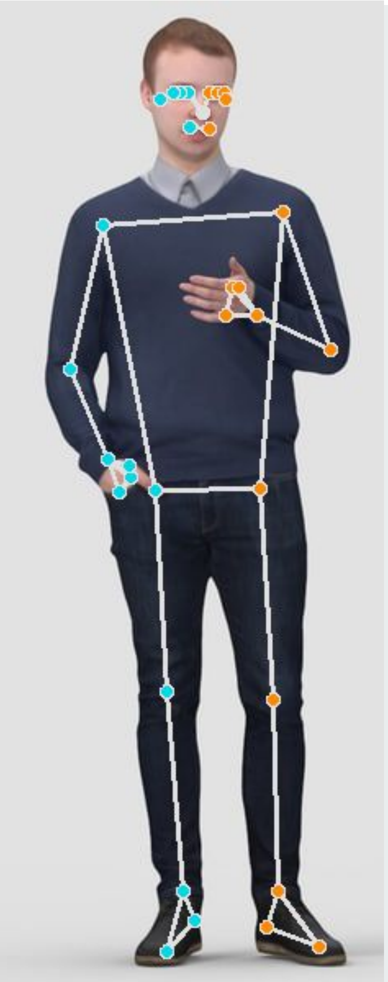
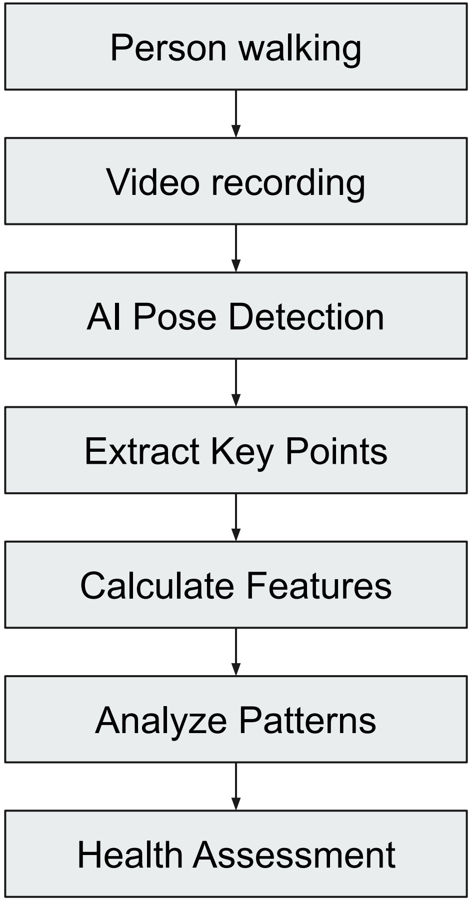


Assessing Health Conditions with Machine Learning-based Gait Analysis

ASDRP
Alex Mui
... and team

Project Abstract

Many brain and muscle disorders change gait (the way people walk), but gait exams often require clinics or specialized equipment. This project tests whether Youtube quality walking video can be analyzed with pose estimation and machine learning to sort walking patterns into health groups. For my experiments, I was able to find the recently released Gait Abnormality in Video Dataset (GAVD) dataset with online links to gait videos with clinical annotations. I used Google's MediaPipe package to track body landmarks in each video frame, then converted them into quantitative anatomical features: hip, knee, and ankle angles; range of motion; left-right symmetry; step length; step timing; and step-to-step variability. With the GAVD datasets, I trained ML models (k-Nearest Neighbors and Decision Trees) to classify each gait sequence as normal, Parkinsonian, stroke, cerebral palsy, or myopathic. I divided the dataset into 70% training and 30% testing groups. I evaluated the models with class-by-class accuracy. This work affirmatively shows that simple ML techniques can classify short walking videos into health condition groups with > 50% accuracy. More advanced techniques are likely able to improve my results.



Identify the Need for your Prototype

Parkinson's affect many elderly round the world. An easily accessible tool to assess health conditions can help alert family members earlier so that they can more readily help take care of affected elderly. The **scope** is huge:

- 120,000 people die from Parkinson's every year
- This project also aims to provide easily accessible tools (a phone or computer with webcam) to be used by anyone to assess health conditions based on short walking videos

Identify the Need for your Prototype

- Stride Length <1.0 m
- Walking Speed <1.0 m/s
- Increased Cadence >120 steps/min
- Reduced arm swing
- Increased double support time >25%
- Forward trunk lean >8°
- Shuffling (reduced ground clearance)

These are all signs in my grandfather's gait that very clearly show that he had Parkinson's. I wonder…

Research Question

Can a computer vision model accurately classify different health conditions using only a short walking video?

Current Research

Recent clinical research shows that markerless, video-based pose estimation can extract high quality gait measures from a single camera. For example, a large open-access study led by ***Stenum et al*** (2024) showed that software inferred keypoints from simple tablet videos can be used to produce quantitative gait measurements in clinical groups such as post-stroke and Parkinson's, which suggests that low-cost video can capture real gait differences. ***Sato, et al*** (2019) used home videos and pose estimation software to measure gait features that separated normal and Parkinsonian walking, showing that such non-professionally taken videos are still useful for diagnosing clinical conditions.

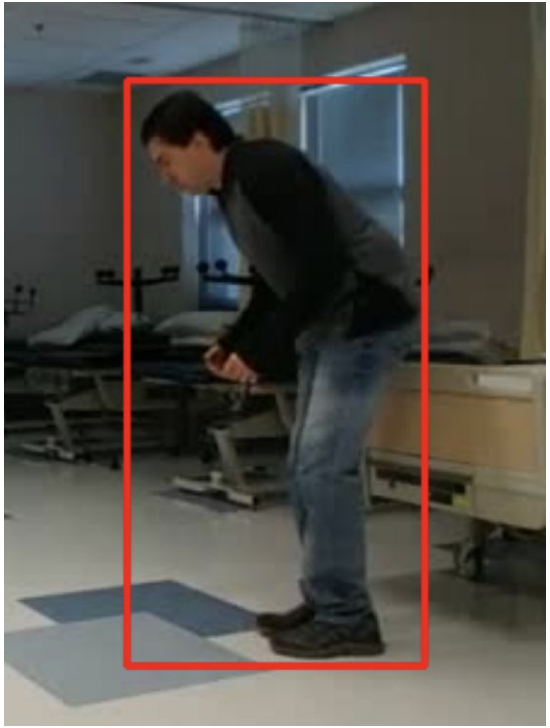
Rangan, et al (2024) recently released the Gait Abnormality in Video Dataset (GAVD) which is now the largest collection of online links to gait videos with clinical annotations. This dataset made possible more rigorous training and fine-tuning of Clinical Gait Analysis using computer vision models, such as those used in this research.

Recent validation work by ***John, et al*** (2024) asked: are these video measurements accurate enough to be trusted? Open-access results show pose-estimation gait measures can be reasonably accurate for step timing and some joint motions, especially when the camera view is chosen carefully. Studies by ***Samadi Kohnehshahri, et al*** (2024) in stroke and cerebral palsy report that machine learning can classify gait patterns, but performance often drops when datasets are small or when videos come from different cameras and settings. In our experiments, we can confirm the scale challenge for real-world screening. Even with the GAVD dataset, we had only 68 relevant videos for the diseases that we studied in this research.

GAVD Dataset

There's a few reasons I chose the Gait Abnormality in Video Dataset (GAVD) dataset over others for my research:

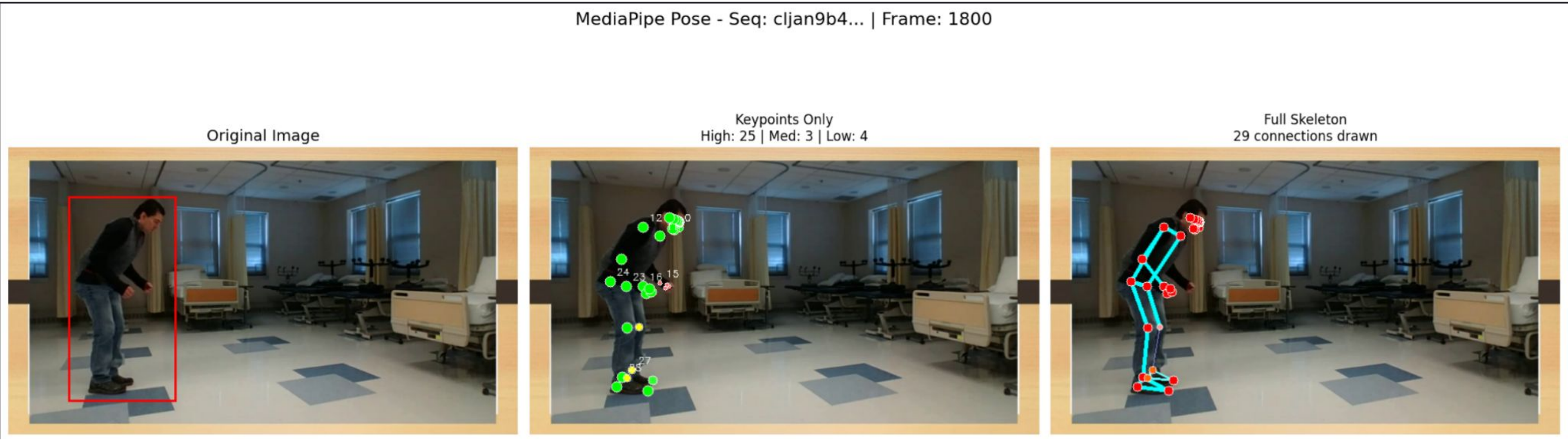
- As of 2025, GAVD is the largest collection of thousands of online links to gait videos with clinical annotations, and it is freely available
- The dataset is designed for Clinical Gait Analysis using computer vision however has many applications such as gait recognition and abnormality action detection.
- It is very high quality – it was created by a PhD student (now Dr. Rahm Rangan) in Australia. Dr. Rangan organized GAVD in a way very easy for me, the human, and the computer to interpret and use
- The dataset contains 'bounding boxes' around all walking subjects in each video — very helpful in this project specifically



Prototype

MediaPipe

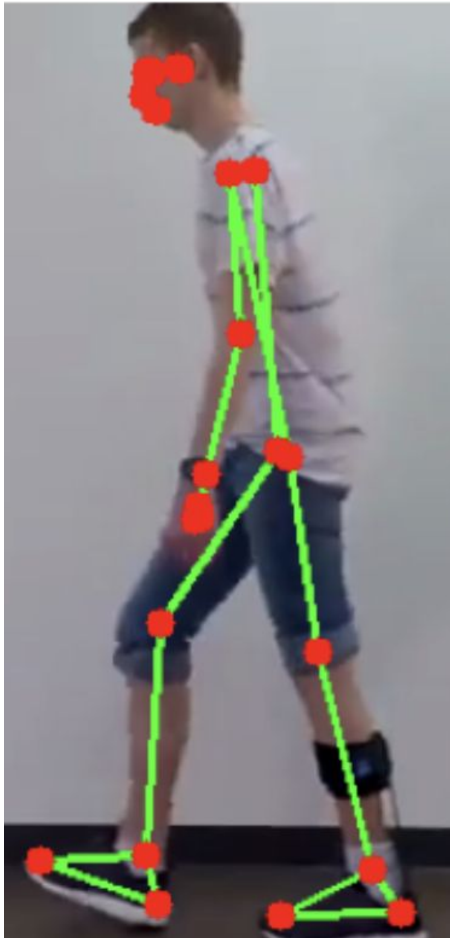
- Used Google's **MediaPipe** library, an open source project, to detect landmarks on human bodies in images or videos.
- MediaPipe can help identify important locations ("**keypoints**") in a body and categorize movements.
- It outputs body pose landmarks using **BLAZEPOSE_33**, a format of 33 known keypoints, or landmarks, in someone's body, including knee joints, eyes, and elbow joints.
- The landmarks are created in **3-D world coordinates** that can be used to align to real-world length and velocity measurements, but is later converted to 2D video.



Prototype

```
Total features loaded: 68

Condition distribution:
  cerebralpalsy   :  15 samples
  myopathic       :  20 samples
  normal          :  12 samples
  parkinsons      :   9 samples
  stroke          :  12 samples
```



Design criteria

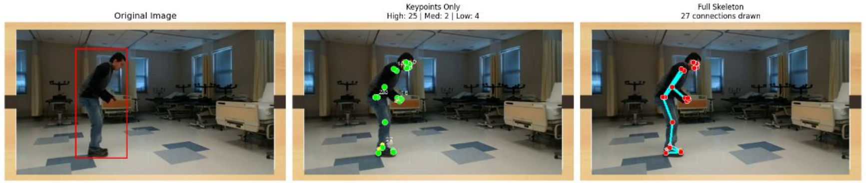
- Use **MediaPipe** BLAZEPOSE-33 keypoints
- **Measuring the accuracy of classification** of each gait sequence into 1 of 5 health conditions
- Extracted features (step time/length, symmetry/asymmetry, joint angles)
- **Evaluation** using a 70% train / 30% test split on 68 clinically annotated sequences (GAVD) and reporting accuracy as metric

Alternate designs considered

- Different **feature sets for classification**:
  - Joint angles, spatiotemporal, symmetry, kinematics, variabilities
- Different **ML classifiers & parameters** (all using scikit-learn library):
  - k-Nearest Neighbors with (k = **3**, 4, 5)
  - Decision Trees with (depth = 3, 4, **5**)
  - Random Forest with (n_estimators = 50, **100**, 200)
- Different **validation** strategies: different splitting (50/50, 60/40, **70/30,** 80/20)

Prototype

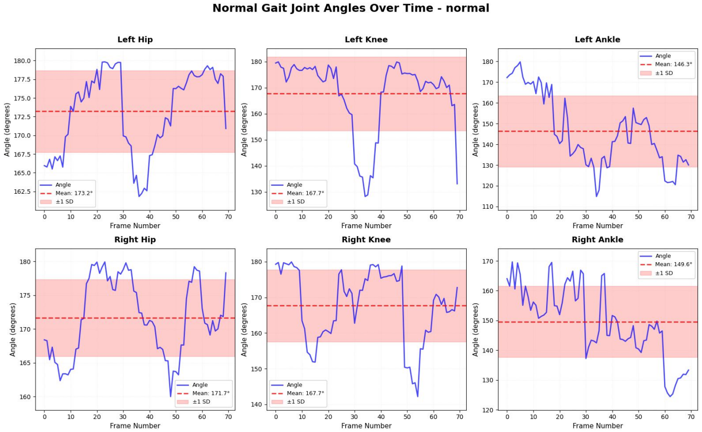
Extracting keypoints




```
Detailed Pose Analysis:
    Total landmarks: 33
    Visible landmarks (>0.5): 25
    High confidence (>0.7): 25
    Medium confidence (0.4-0.7): 2
    Low confidence (0.2-0.4): 4
    Average confidence: 0.767
    Skeleton connections drawn: 27

Body Part Confidence:
    Face (0-10): 0.989
    Upper body (11-22): 0.562
    Lower body (23-32): 0.767
```

Joint angle features



Extracting features

Gait Classifier

1. Load extracted features

In [1]:

```python
from ambient.classification.features import GaitFeatureVector
from ambient.utils.features_utils import load_features

FILENAME="enhanced_features"
DIRECTORY="features"

all_features, condition_counts = load_features(filename=FILENAME, directory=DIRECTORY)
print(f"\n{'='*60}")
print(f"Total features loaded: {len(all_features)}")
print(f"\nCondition distribution:")
for condition, count in sorted(condition_counts.items()):
    print(f"  {condition:15s}: {count:3d} samples")
```

```
✓ Features loaded successfully from: features/enhanced_features
  Saved on: 2026-01-25T00:49:49.819786
  Total features: 68
  Conditions: ['stroke', 'myopathic', 'parkinsons', 'normal', 'cerebralpalsy']

============================================================
Total features loaded: 68

Condition distribution:
  cerebralpalsy  :  15 samples
  myopathic      :  20 samples
  normal         :  12 samples
  parkinsons     :   9 samples
  stroke         :  12 samples
```

Classifying features

In [45]:

```python
report = eval_metrics['classification_report']
for class_name in eval_metrics['classes']:
    cm = report[class_name]
    print(f"\n{class_name}")
    print(f"  F1-Score : {cm['f1-score']:.3f}")
    print(f"  Precision: {cm['precision']:.3f}")
    print(f"  Recall   : {cm['recall']:.3f}")
```

```
cerebralpalsy
  F1-Score : 0.462
  Precision: 0.375
  Recall   : 0.600

myopathic
  F1-Score : 0.750
  Precision: 0.667
  Recall   : 0.857

normal
  F1-Score : 0.667
  Precision: 0.667
  Recall   : 0.667

parkinsons
  F1-Score : 0.000
  Precision: 0.000
  Recall   : 0.000

stroke
  F1-Score : 0.000
  Precision: 0.000
  Recall   : 0.000
```

The Design

Design goal: Use pose estimation + machine learning in a Jupyter notebook to classify gait as Parkinson's, stroke, cerebral palsy, myopathic, or normal.

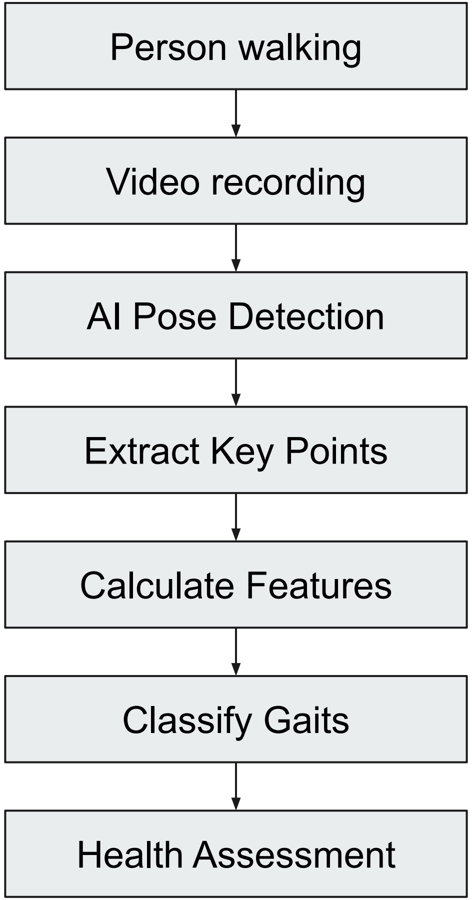
How the prototype works

Input data: 68 clinically annotated gait videos (GAVD)

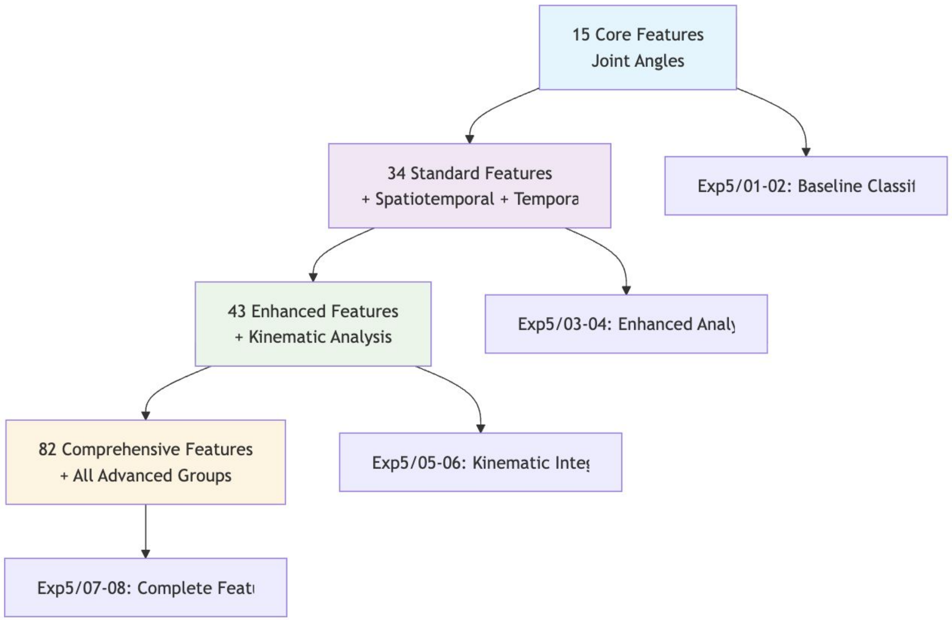
1. **Pose estimation:** MediaPipe tracks body **keypoints in every frame**
2. **Feature building:** Convert keypoints → gait features (examples):
    - hip/knee/ankle angles + range of motion
    - step time & step length
    - left–right asymmetry + step-to-step consistency
3. **Model training/testing:** 70% train / 30% test using
 k-NN (k=3), Decision Tree (depth=5), Random Forest (n=100)
4. **Evaluation criteria:** select the classifier with the highest accuracy for the test set

Main constraints

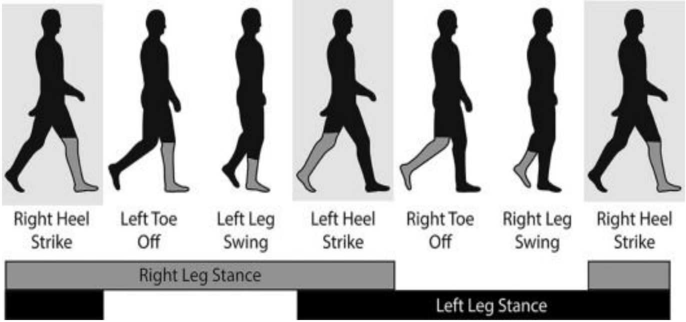
- **Small dataset** (68 videos) limits how well results generalize
- 2D pose + camera angle/occlusion can reduce keypoint accuracy

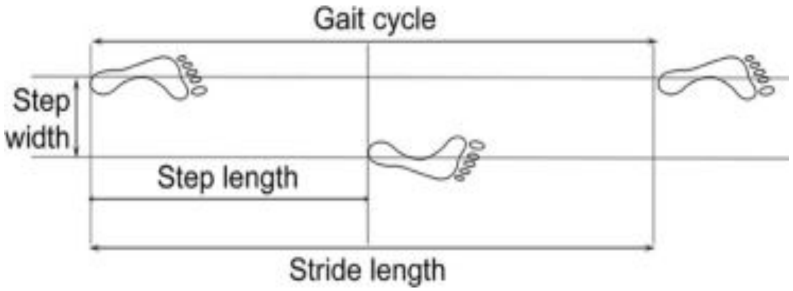


Gait Features for Extraction & Classification



Gait Features for Extraction & Classification

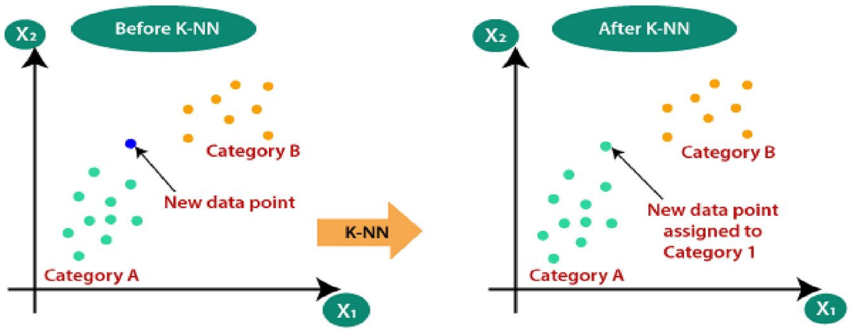


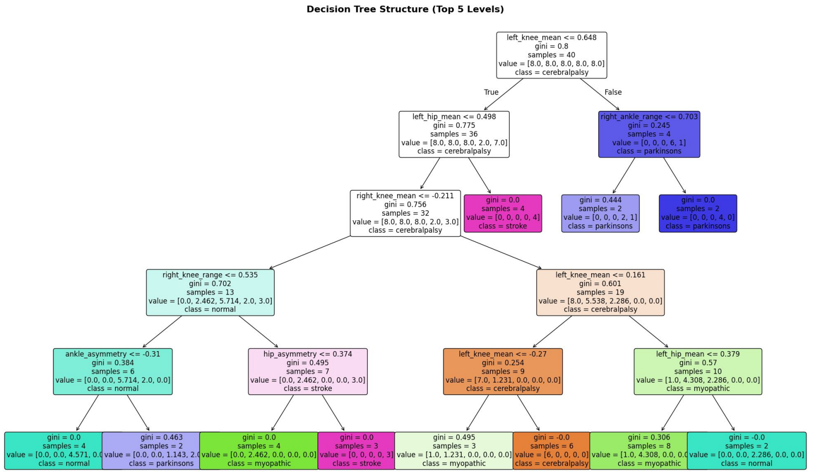


Joint Angles	Spatiotemporal	Symmetry	Kinematics	Variability	Temporal Extended
left_hip_mean	walking_speed_ms	stride_length_si	velocity_mean	stride_time_cv	stance_percentage
left_knee_mean	cadence_steps_min	stance_time_si	velocity_std	step_length_cv	swing_percentage
left_ankle_mean	stride_length_m	swing_time_si	velocity_max	stride_velocity_cv	double_support_percent
right_hip_mean	step_width_m	hip_angle_si	velocity_min		stance_swing_ratio
right_knee_mean		knee_angle_si	acceleration_mean		sequence_length
right_ankle_mean		ankle_angle_si	acceleration_std	**Symmetry Extended**	duration_seconds
left_hip_range			acceleration_max	shoulder_symmetry_index	dominant_frequency
left_knee_range	**Spatial**	**Temporal**	jerk_mean	elbow_symmetry_index	cycle_count
left_ankle_range	step_width_std,	walking_speed_ms	jerk_std	wrist_symmetry_index	left_cycle_duration_mean
right_hip_range	step_width_range,	cadence_steps_min	walking_speed_pixels_per_sec	hip_symmetry_index	right_cycle_duration_mean
right_knee_range	left_ankle_total_distance,	stride_length_m	estimated_stride_length_pixels,	knee_symmetry_index	cycle_duration_asymmetry
right_ankle_range	right_ankle_total_distance,	step_width_m		ankle_symmetry_index	double_support_duration_mean
	ankle_distance_asymmetry,	stance_percentage	**Symmetry**	overall_symmetry_index	stance_duration_mean
	trunk_lean_angle	swing_percentage	hip_asymmetry	positional_symmetry_score	swing_duration_mean
	pelvic_tilt_mean	double_support_percent	knee_asymmetry	movement_symmetry_score	phase_asymmetry
		stance_swing_ratio	ankle_asymmetry	temporal_symmetry_score	fps

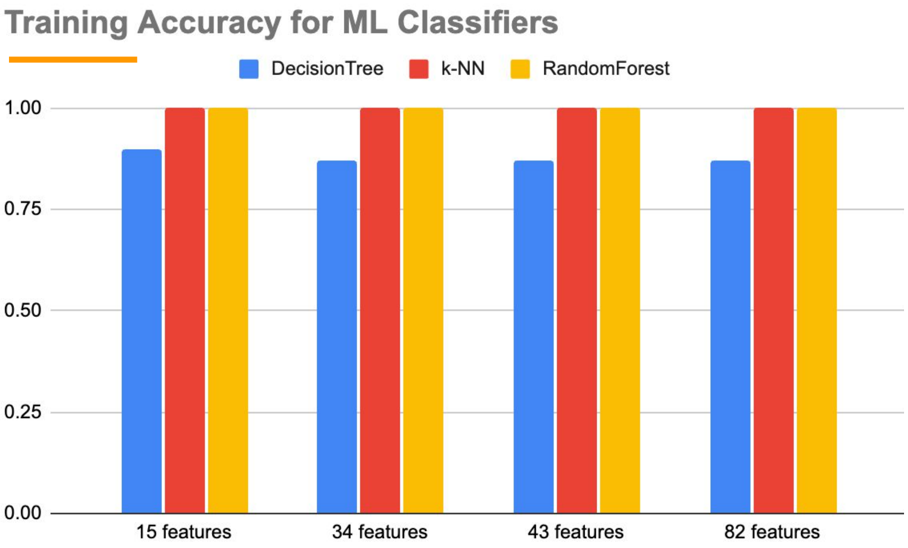
Test Plan

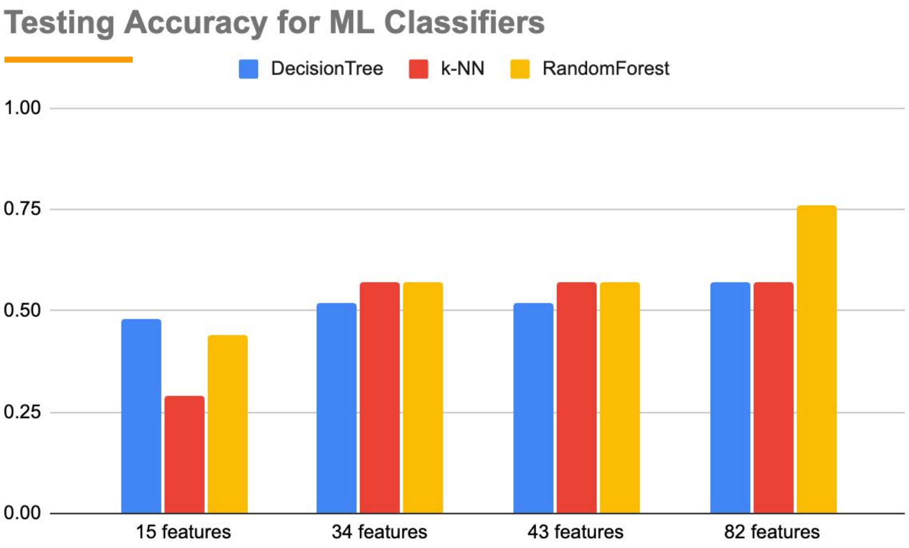
To test the prototype's accuracy, I needed to download clinically accurate gait videos. For my research, I was able to find a well clinically annotated GAVD dataset, a free collection of 68 clinically annotated gait patterns – identified with normal/abnormality labels. I used 3 simple machine learning classifiers to train and test. I tested the k-NN classifier (k=3), the Decision Tree classifier (depth=5), Random Forest (n = 100), all available from the *scikit-learn* library.





Test Results





15 features: (hip, knee, ankle) joint angles, symmetry
34 features: + spatiotemporal (walking speed, cadence, stride length, step width), more symmetry
43 features: + kinematic (velocity, acceleration, jerk)
82 features: + variability (coefficient of variability for stride length, step length, velocity), more temporal and stability features

Analysis of Test Results

How I tested

- Dataset: **68 labeled gait videos (GAVD)**
- Split: **70% training / 30% testing**
- Models compared: **k-NN, Decision Tree, Random Forest**
- Goal: predict gait type (**Parkinson's, stroke, cerebral palsy, myopathic, normal**)

What the results showed

- All models looked good on **training** data, but the real test was **unseen test videos**
- **Random Forest performed best** on the test set
  - Best test accuracy: **0.76** (by using **82 features**)
- **Decision Tree performed worst** on the test set

Lessons learned

- **Random Forest** was the most accurate model we tested
  - More advanced algorithms would likely have better accuracy
- **More labeled data needed:**
  - We needed a lot more labeled training & testing datasets to improve generalization
  - 68 clinically labeled datasets is not enough
- **Some videos are shaky** that led to lower classification accuracy
  - Some smoothing techniques would have helped

Conclusions

Creating a classifier has been both an educational and a fulfilling journey. For my research question:

Can a computer vision model accurately classify different health conditions using only a short walking video?

My research's answer is nuanced: it depends. After trying different types of ML classifiers, the highest accuracy I could achieve was 76% on the test dataset using Random Forest classifier. That's still much higher than choosing a class at random (which would be 20% accuracy because there are five classes). But, it's nowhere near the accuracy it should be at to be useful to the general public. Because I only had 68 datasets to work with, however, I could have searched other data collections for relevant data points to drastically increase the accuracy. Another method I could have used to increase classification accuracy was to use more sophisticated machine learning classification algorithms.

I look forward to continuing this research project in high school, and learning more advanced ML techniques applying to more diverse and challenging real world data problems.

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